When was the First Manmade Fiber Made?

Rayon, was the first man-made fiber created in 1910 and it was called 'artificial silk'. Viscose is the most common form of Rayon.



Rayon Threads

3.2.1 Types of Rayon

Based on the method of manufacture, at present there are two principal types of rayon.

1. Rayon or Viscose
2. High Wet Modulus Rayon (regenerated rayon)

In High Wet Modulus Rayon (HWM), the original material (cellulose) is changed chemically into another form, which is then changed or regenerated into cellulose again. These changes produce the final product that is purified cellulose in the fibre form.

3.2.2 Manufacturing Process of Viscose Rayon

For making viscose rayon, wood chips or cotton linters are treated with chemicals to produce sheets of purified cellulose that resembles white blotters. This purified cellulose is soaked in caustic soda which produces sheets of alkali cellulose. These sheets are then broken up into fluffy white flakes or grains called cellulose crumbs. The crumbs are aged for 2-3 days under controlled temperature and humidity. Liquid carbon disulphide is then added to these cellulose crumbs which turn it into a light orange substance called cellulose xanthate. The cellulose xanthate crumbs are then dissolved in a weak solution of caustic soda, which turns it to a thick viscous solution resembling honey in colour and consistency. This thick solution is called viscose. The viscose is aged, filtered and vacuum treated to remove the air bubbles present, as they may cause the filament to break. This treated viscose solution is then forced through the holes of a spinneret into sulphuric acid which coagulates the cellulose of the cellulose xanthate to form pure regenerated cellulose filaments (Figure 3.1).

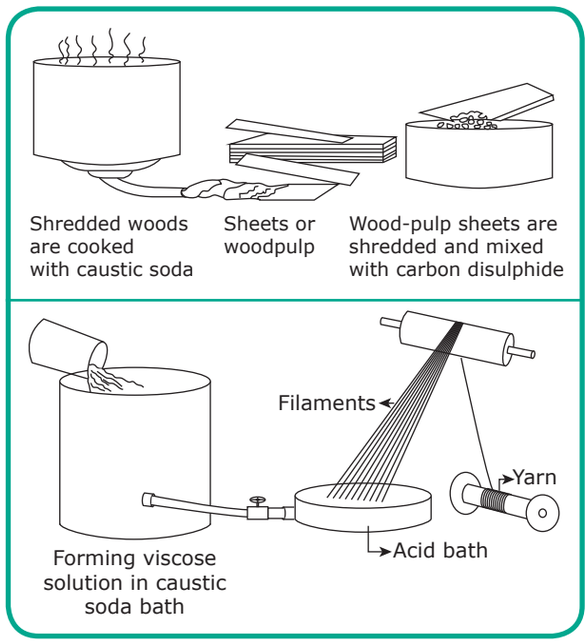


Figure 3.1 Manufacturing Process of Viscose Rayon

3.2.3 Finishing Process

A wide variety of fabrics can be produced by viscose rayon. Spun rayon fabrics can be used for making fabrics that resemble cotton, linen or wool. Rayon filament yarns can be used for making fabric that resemble silk. Various finishes can be

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